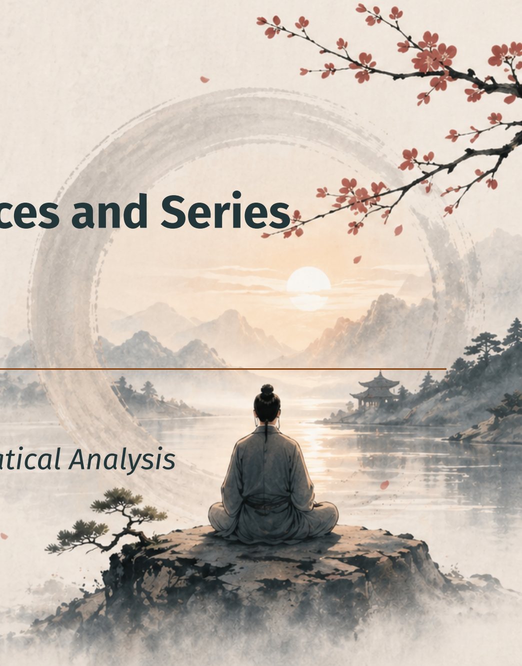
Chapter 3: Numerical Sequences and Series

Section: Series



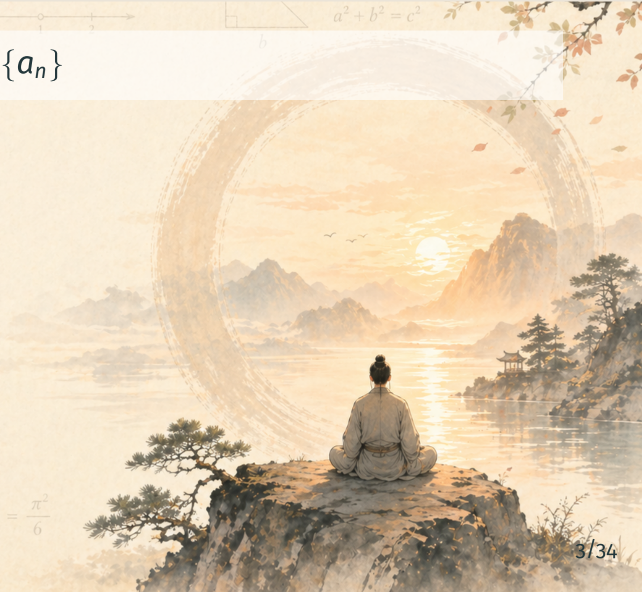
Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

From Sequences to Series

- So far: theory of **sequences** $\{a_n\}$



From Sequences to Series

- So far: theory of sequences $\{a_n\}$
- Yet many natural quantities are *infinite sums*:

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots, \qquad 1 - \frac{1}{3} + \frac{1}{5} - \cdots$$

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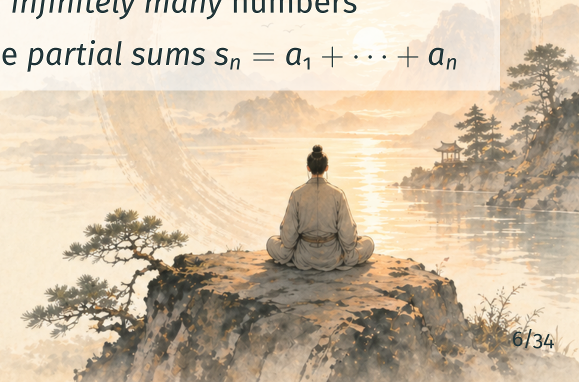
- Problem: we cannot literally add *infinitely many* numbers
- Idea: reduce to sequences via the *partial sums* $s_n = a_1 + \cdots + a_n$



Series and Partial Sums

A Standing Convention for the Rest of the Chapter

Convention

In the remainder of Chapter 3, all sequences and series are complex-valued unless explicitly stated otherwise.

Extensions to $\mathbb{R}^k$-valued series are treated in Exercise 15.

Definition 3.21: Finite Sum Notation

Finite sums: the Σ symbol

Given a sequence $\{a_n\}$ and integers $p \le q$,

$$\sum_{n=p}^{q} a_n = a_p + a_{p+1} + \cdots + a_q.$$

Capital sigma, Σ, is just compact notation for a finite sum — nothing infinite yet.

Definition 3.21: Partial Sums and the Series Symbol

Partial sums and infinite series

With each sequence $\{a_n\}$ we associate the sequence of partial sums

$$s_n = \sum_{k=1}^{n} a_k = a_1 + a_2 + \cdots + a_n.$$

The symbolic expression

$$a_1 + a_2 + a_3 + \cdots \qquad \text{or, more concisely,} \qquad \sum_{n=1}^{\infty} a_n \tag{4}$$

is called an *infinite series*, or just a *series*.

Definition 3.21: Convergence and the Sum

Convergence, divergence, sum

If the partial sums $\{s_n\}$ converge to s, the series converges and we write

$$\sum_{n=1}^{\infty} a_n = s.$$

The number s is the *sum of the series*.

If $\{s_n\}$ diverges, the series diverges.

Crucial caveat

s is the limit of a sequence of sums, not a sum performed in one step.

Definition 3.21: Notational Variations

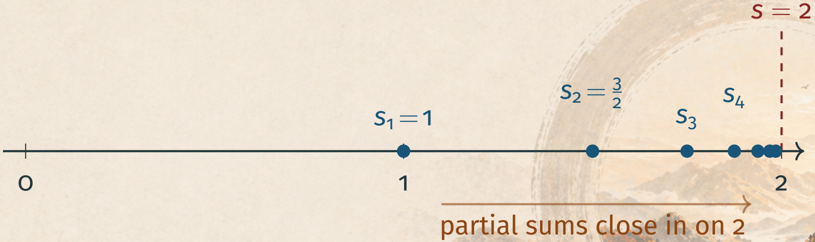
Two notational shortcuts

- Sometimes we start at $n = 0$: $\displaystyle\sum_{n=0}^{\infty} a_n$.
- When the starting index does not matter, we just write $\sum a_n$.

Sequences and series are interchangeable:

Set $a_1 = s_1,\ a_n = s_n - s_{n-1}\ (n > 1)$ to go back from a sequence $\{s_n\}$ to its generating series.

Example: The Partial Sums of $1 + \frac{1}{2} + \frac{1}{4} + \cdots$



Each s_n shrinks the remaining gap by a factor of 2.

The Cauchy Criterion for Series

Theorem 3.22: Cauchy Criterion for Series

Theorem 3.22

$\Sigma\, a_n$ converges if and only if for every $\varepsilon > 0$ there is an integer N such that

$$\left|\sum_{k=n}^{m} a_k\right| \leq \varepsilon \qquad \text{whenever } m \geq n \geq N. \tag{6}$$

Idea: apply the Cauchy criterion for *sequences* (Theorem 3.11) to the partial sums $\{s_n\}$.

Why Theorem 3.22 Follows from Theorem 3.11

The translation in one identity

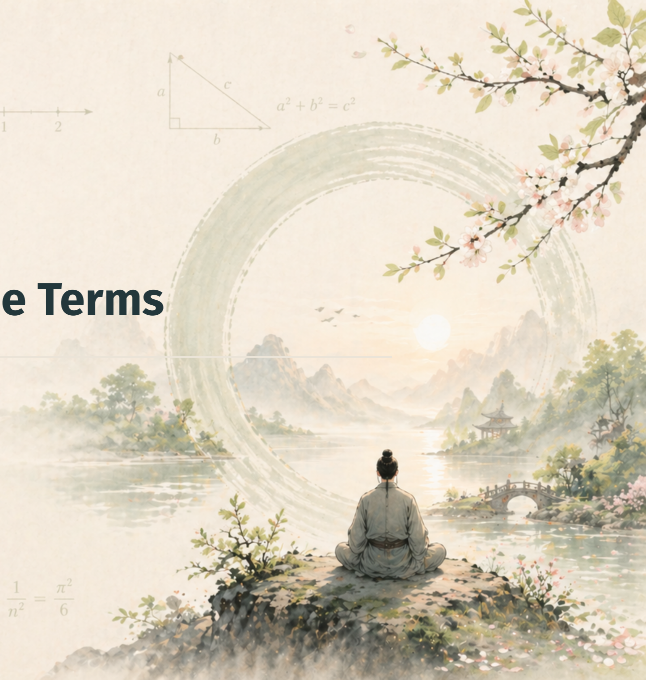
For $m \geq n \geq 1$,

$$s_m - s_{n-1} = \sum_{k=n}^{m} a_k.$$

So a block sum on the right is the difference of two partial sums.

Therefore: $\left|\sum_{k=n}^{m} a_k\right| \leq \varepsilon$ for all $m \geq n \geq N$
$\iff$ $|s_m - s_{n-1}| \leq \varepsilon$ for all $m \geq n \geq N$
$\iff$ $\{s_n\}$ is Cauchy $\iff$ $\{s_n\}$ converges (Theorem 3.11)
$\iff$ $\sum a_n$ converges.

Vanishing of the Terms



Theorem 3.23: A Necessary Condition for Convergence

Theorem 3.23

If $\sum a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

Why: take $m = n$ in Theorem 3.22

With $m = n$, the block sum is just the single term a_n, so (6) becomes

$$|a_n| \le \varepsilon \qquad (n \ge N).$$

Hence $a_n \to 0$. $\square$

Warning: The Converse Fails

Theorem 3.23 is necessary, not sufficient

$a_n \to 0$ does not imply $\sum a_n$ converges.

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Standard counterexample: the harmonic series

The series

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

diverges, even though $1/n \to 0$.

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diverges, even though $1/n \to 0$.

The terms going to zero says *nothing* about whether the sum is finite.

Series of Nonnegative Terms

Theorem 3.24: Nonnegative Series and Bounded Partial Sums

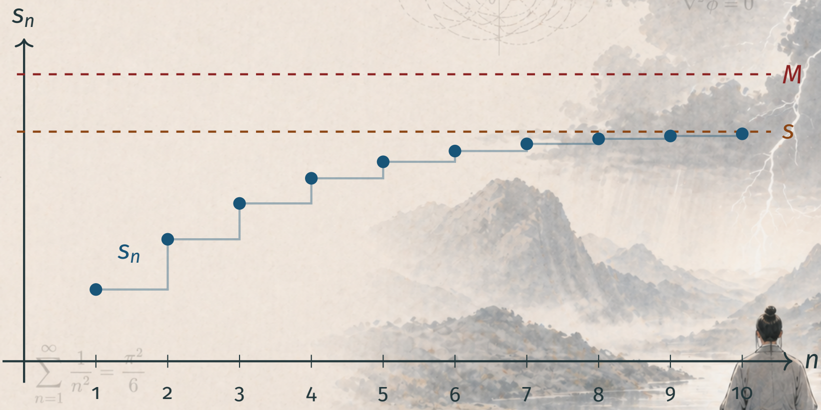
Theorem 3.24

A series of nonnegative real terms converges if and only if its partial sums form a bounded sequence.

Idea: when $a_n \geq 0$, the partial sums $s_n = a_1 + \cdots + a_n$ are monotonic.

Apply Theorem 3.14: monotonic + bounded $\Longrightarrow$ convergent.

Picture: A Monotone Staircase of Partial Sums



Increasing + bounded by $M \Longrightarrow$ a finite limit $s \leq M$.

Why Theorem 3.24 Matters in Practice

Convergence reduces to a single quantitative question:

- Is there a constant M with $s_n = a_1 + a_2 + \cdots + a_n \leq M$ for all n?
- If yes, $\sum a_n$ converges (to some $s \leq M$).
- If no, $\sum a_n$ diverges to $+\infty$.

This is what makes the comparison test coming up so powerful.

The Comparison Test

Theorem 3.25: The Comparison Test

Theorem 3.25 (Comparison test)

(a) If $|a_n| \leq c_n$ for $n \geq N_0$ and $\sum c_n$ converges, then $\sum a_n$ converges.

(b) If $a_n \geq d_n \geq 0$ for $n \geq N_0$ and $\sum d_n$ diverges, then $\sum a_n$ diverges.

Caveat

Part (b) applies only to series of nonnegative terms a_n.

Proof of Theorem 3.25 (a) — Setup

Step 1: choose N using Cauchy on $\sum c_n$

Since $\sum c_n$ converges, by Theorem 3.22, given $\varepsilon > 0$ there exists $N \geq N_0$ such that

$$\sum_{k=n}^{m} c_k \leq \varepsilon \qquad \text{whenever } m \geq n \geq N.$$

Proof of Theorem 3.25 (a) — The Estimate

Step 2: triangle inequality + comparison

For $m \geq n \geq N$:

$$\left|\sum_{k=n}^{m} a_k\right| \;\leq\; \sum_{k=n}^{m} |a_k| \;\leq\; \sum_{k=n}^{m} c_k \;\leq\; \varepsilon.$$

Three moves: triangle inequality, term-by-term comparison $|a_k| \leq c_k$, then Cauchy on $\sum c_n$.

Proof of Theorem 3.25 (a) — Conclusion

Step 3: Cauchy on $\sum a_n$

We have shown: for every $\varepsilon > 0$ there is N with

$$\left|\sum_{k=n}^{m} a_k\right| \leq \varepsilon \qquad (m \geq n \geq N).$$

By Theorem 3.22, $\sum a_n$ converges. $\square$

Pattern: "Cauchy on the dominating series" $\Longrightarrow$ "Cauchy on the dominated series."

Proof of Theorem 3.25 (b)

Part (b) is just the contrapositive of (a)

Suppose $\sum a_n$ *did* converge, with $0 \le d_n \le a_n$ for $n \ge N_0$.

Then by part (a), $\sum d_n$ also converges — contradicting our hypothesis.

Therefore $\sum a_n$ diverges. $\square$

Alternative

Part (b) also follows directly from Theorem 3.24: the partial sums of $\sum a_n$ dominate the unbounded partial sums of $\sum d_n$, hence are themselves unbounded.

Why the Comparison Test Is So Useful

Strategy for studying a new series $\sum a_n$:

1. Bound $|a_n|$ above by the term c_n of a series known to converge $\Longrightarrow$ convergence.
2. Bound a_n (nonnegative) below by the term d_n of a series known to diverge $\Longrightarrow$ divergence.

Prerequisite: a stockpile of reference series. The next section builds exactly that catalogue (geometric series, p-series, etc.).

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Key takeaways from this section:

- Definition 3.21: a series $\sum a_n$ is just a symbol; its value is $\lim_n s_n$ when this limit exists

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- Theorem 3.23: $\sum a_n$ converges $\implies a_n \to 0$ *(converse fails!)*
- Theorem 3.24: if $a_n \geq 0$, convergence $\iff$ partial sums bounded
- Theorem 3.25 (Comparison test): dominate by a known convergent / divergent series
- *Slogan:* a series is its sequence of partial sums

Coming Up Next

Next section: Series of Nonnegative Terms

- Build the reference catalogue the comparison test depends on
- Geometric series, the harmonic series, p-series
- The Cauchy condensation test

Together these turn Theorem 3.25 into a practical convergence test for almost everything you will meet.